NanoGT Match Report: Friedreich ataxia

Disease: Friedreich ataxia (ORPHA:95)
Primary gene: FXN
Gene CDS: 633 bp
Inheritance: Autosomal recessive
Target tissues scored: CNS, heart

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: MITOCHONDRIAL MATRIX ENZYME (FXN): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.; Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints; Cell-autonomous protein across multiple tissues; high transduction coverage may be required.

Disease Mechanism Evidence

Molecular mechanism: repeat expansion silencing
Mechanistic detail: GAA trinucleotide repeat expansion in FXN intron 1 induces heterochromatin formation with H3K9me3 and DNA hypermethylation silencing frataxin transcription; the FXN protein coding sequence is structurally normal so a transgenic cDNA (without the expanded intron) can restore frataxin expression
Gene-addition compatibility: conditional
Preferred modality class: frataxin gene addition or epigenetic derepression
Evidence level/status: direct / source_checked
Evidence summary: Campuzano et al. (1996 Science PMID 8596916) identified the intronic GAA expansion; Al-Mahdawi et al. (2008 Hum Mol Genet PMID 18045775) characterised the heterochromatin mechanism; FXN cDNA gene addition is actively pursued clinically because the normal frataxin protein sequence is preserved
Evidence source: Campuzano et al. 1996 Science PMID 8596916; Al-Mahdawi et al. 2008 Hum Mol Genet PMID 18045775

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
- Endpoint risk: muscle/cardiac diseases may need functional, respiratory, imaging, or cardiac endpoints that progress slowly and vary by age/stage.
- Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	OAV101-IT	AAV9	7.9/10	High	approved
2	Zolgensma	AAV9	7.9/10	High	approved
3	SRP-9001	AAV9	7.0/10	Medium	approved
4	BMN 307	AAV5	7.0/10	Medium	phase2
5	Libmeldy	LV	7.0/10	Medium	approved

Match #1: OAV101-IT

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/spinal cord
Composite score: 7.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment

Dimension	Score	Max	What it measures
TOTAL (normalised)	**7.90**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 633bp / cargo 4700bp (13% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: motor_neuron
- Disease mechanism: repeat expansion silencing — GAA trinucleotide repeat expansion in FXN intron 1 induces heterochromatin formation with H3K9me3 and DNA hypermethylation silencing frataxin transcription; the FXN protein coding sequence is structurally normal so a transgenic cDNA (without the expanded intron) can restore frataxin expression
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Campuzano et al. (1996 Science PMID 8596916) identified the intronic GAA expansion; Al-Mahdawi et al. (2008 Hum Mol Genet PMID 18045775) characterised the heterochromatin mechanism; FXN cDNA gene addition is actively pursued clinically because the normal frataxin protein sequence is preserved
- Mechanism source: Campuzano et al. 1996 Science PMID 8596916; Al-Mahdawi et al. 2008 Hum Mol Genet PMID 18045775 (https://pubmed.ncbi.nlm.nih.gov/8596916/)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: CONDITIONAL — FXN is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import efficiency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (FXN): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an

additional disease-specific development step requiring experimental validation.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Zolgensma

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/motor neuron
Composite score: 7.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.90**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 633bp / cargo 4700bp (13% utilized)
- Vector tropism plus precedent target match: cns

- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: motor_neuron
- Disease mechanism: repeat expansion silencing — GAA trinucleotide repeat expansion in FXN intron 1 induces heterochromatin formation with H3K9me3 and DNA hypermethylation silencing frataxin transcription; the FXN protein coding sequence is structurally normal so a transgenic cDNA (without the expanded intron) can restore frataxin expression
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Campuzano et al. (1996 Science PMID 8596916) identified the intronic GAA expansion; Al-Mahdawi et al. (2008 Hum Mol Genet PMID 18045775) characterised the heterochromatin mechanism; FXN cDNA gene addition is actively pursued clinically because the normal frataxin protein sequence is preserved
- Mechanism source: Campuzano et al. 1996 Science PMID 8596916; Al-Mahdawi et al. 2008 Hum Mol Genet PMID 18045775 (https://pubmed.ncbi.nlm.nih.gov/8596916/)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: CONDITIONAL — FXN is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import efficiency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (FXN): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: SRP-9001

Precedent disease: Duchenne muscular dystrophy
Vector: AAV9
Tissue target: muscle
Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.05**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 633bp / cargo 4700bp (13% utilized)
- Vector tropism overlap: cns, heart
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Pathway match: myopathy
- Disease mechanism: repeat expansion silencing — GAA trinucleotide repeat expansion in FXN intron 1 induces heterochromatin formation with H3K9me3 and DNA hypermethylation silencing frataxin transcription; the FXN protein coding sequence is structurally normal so a transgenic cDNA (without the expanded intron) can restore frataxin expression

- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Campuzano et al. (1996 Science PMID 8596916) identified the intronic GAA expansion; Al-Mahdawi et al. (2008 Hum Mol Genet PMID 18045775) characterised the heterochromatin mechanism; FXN cDNA gene addition is actively pursued clinically because the normal frataxin protein sequence is preserved
- Mechanism source: Campuzano et al. 1996 Science PMID 8596916; Al-Mahdawi et al. 2008 Hum Mol Genet PMID 18045775 (https://pubmed.ncbi.nlm.nih.gov/8596916/)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: CONDITIONAL — FXN is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import efficiency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (FXN): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: BMN 307

Precedent disease: Phenylketonuria
Vector: AAV5
Tissue target: liver
Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.00**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 633bp / cargo 4700bp (13% utilized)
- Vector tropism overlaps cns, but precedent target is liver
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (myopathy vs amino_acid_metabolism)
- Disease mechanism: repeat expansion silencing — GAA trinucleotide repeat expansion in FXN intron 1 induces heterochromatin formation with H3K9me3 and DNA hypermethylation silencing frataxin transcription; the FXN protein coding sequence is structurally normal so a transgenic cDNA (without the expanded intron) can restore frataxin expression
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Campuzano et al. (1996 Science PMID 8596916) identified the intronic GAA expansion; Al-Mahdawi et al. (2008 Hum Mol Genet PMID 18045775) characterised the heterochromatin mechanism; FXN cDNA gene addition is actively pursued clinically because the normal frataxin protein sequence is preserved
- Mechanism source: Campuzano et al. 1996 Science PMID 8596916; Al-Mahdawi et al. 2008 Hum Mol Genet PMID 18045775 (https://pubmed.ncbi.nlm.nih.gov/8596916/)
- Approval status: phase2

- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: CONDITIONAL — FXN is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import efficiency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (FXN): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: heart
- Only partial tissue match; verify target-cell transduction and delivery route manually
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: Libmeldy

Precedent disease: Metachromatic leukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**6.95**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 633bp / cargo 8000bp (8% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (myopathy vs leukodystrophy)
- Disease mechanism: repeat expansion silencing — GAA trinucleotide repeat expansion in FXN intron 1 induces heterochromatin formation with H3K9me3 and DNA hypermethylation silencing frataxin transcription; the FXN protein coding sequence is structurally normal so a transgenic cDNA (without the expanded intron) can restore frataxin expression
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Campuzano et al. (1996 Science PMID 8596916) identified the intronic GAA expansion; Al-Mahdawi et al. (2008 Hum Mol Genet PMID 18045775) characterised the heterochromatin mechanism; FXN cDNA gene addition is actively pursued clinically because the normal frataxin protein sequence is preserved
- Mechanism source: Campuzano et al. 1996 Science PMID 8596916; Al-Mahdawi et al. 2008 Hum Mol Genet PMID 18045775 (https://pubmed.ncbi.nlm.nih.gov/8596916/)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial

- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: CONDITIONAL — FXN is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import efficiency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (FXN): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, heart
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable